

MIAE 221: Materials Science for Engineers

Chapter 3: The Structure of Crystalline Solids (Expanded & Condensed Guide)

Concordia University · Gina Cody School of Engineering

Textbook:

Materials Science and Engineering: An Introduction

(10th Ed., Callister & Rethwisch)

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(Aligned with Dr. Medraj Lectures 4, 5 & 6 · Weeks 2–3)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Atoms in solids don't arrange randomly (unless they are glass or polymers). Metals arrange in tightly ordered 3D repeating grids called **crystal lattices**. How tightly and efficiently atoms pack determines how dense the metal is, how easily it deforms under a hammer, and how it diffracts X-rays.

Real-World Analogy

: Stacking oranges in a grocery store display. If you stack them directly on top of each other in a grid, you get Simple Cubic (very loose, wobbles easily). If you drop each orange into the valley between oranges in the layer below, you get close-packing (FCC or HCP, rock solid, maximum packing density).

2. ⚙️ High-Yield Mathematical Engine & Metallic Unit Cells

Crystal Structure	Coordination Number (CN)	Atoms per Cell (n)	Lattice Parameter $a(R)$	Atomic Packing Factor (APF)	Close-Packed Direction
Simple Cubic (SC)	6	1	$a = 2R$	$\frac{\pi}{6} \approx 0.524$	$\langle 100 \rangle$
Body-Centered Cubic (BCC)	8	2	$a = \frac{4R}{\sqrt{3}}$	$\frac{\pi\sqrt{3}}{8} \approx 0.680$	$\langle 111 \rangle$
Face-Centered Cubic (FCC)	12	4	$a = 2\sqrt{2}R$	$\frac{\pi\sqrt{2}}{6} \approx 0.740$	$\langle 110 \rangle$
Hexagonal Close-Packed (HCP)	12	6	$a = 2R, c = 1.633a$	$\frac{\pi\sqrt{3}}{8} \approx 0.740$	$\langle 11\bar{2}0 \rangle$

Essential Crystallographic Equations:

1. Theoretical Density:

$$\rho = \frac{n \cdot A}{V_c \cdot N_A}$$

(Units: A in g/mol, V_c in cm^3 , $N_A = 6.022 \times 10^{23}$ atoms/mol, yielding ρ in g/cm^3)

2. Linear Density (LD):

$$\text{LD} = \frac{\text{number of atom diameters centered on direction vector}}{\text{length of direction vector}}$$

3. Planar Density (PD):

$$\text{PD} = \frac{\text{number of atom cross-sectional areas centered on plane}}{\text{area of plane}}$$

4. Bragg's Law for X-Ray Diffraction (XRD):

$$n\lambda = 2d_{hkl} \sin \theta$$

Interplanar spacing for cubic systems

:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

5. Diffraction Reflection Selection Rules:

- **BCC**: Reflections occur **only** when $h + k + l = \text{even}$ (e.g., (110), (200), (211), (220)).
- **FCC**: Reflections occur **only** when h, k, l are **unmixed** (all odd or all even) (e.g., (111), (200), (220), (311)).

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

the atomic hard-sphere model) divided by the unit cell volume—that is,

$$\text{APF} = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}} \quad (3.3)$$

For the FCC structure, the atomic packing factor is 0.74, which is the maximum packing possible for spheres all having the same diameter. Computation of this APF is also included as an example problem. Metals typically have relatively large atomic packing factors to maximize the shielding provided by the free electron cloud.

The Body-Centered Cubic Crystal Structure

Another common metallic crystal structure also has a cubic unit cell with atoms located at all eight corners and a single atom at the cube center. This is called a **body-centered cubic (BCC)** crystal structure. A collection of spheres depicting this crystal structure is shown in Figure 3.2c, whereas Figures 3.2a and 3.2b are diagrams of BCC unit cells with

Figure 3.2: Body-Centered Cubic (BCC) unit cell: (a) Hard-sphere model, (b) Reduced-sphere unit cell, and (c) Aggregate of atoms highlighting center-corner contact along the body diagonal.

Figure 3.22 Schematic diagram of an x-ray diffractometer; T = x-ray source, S = specimen, C = detector, and O = the axis around which the specimen and detector rotate.

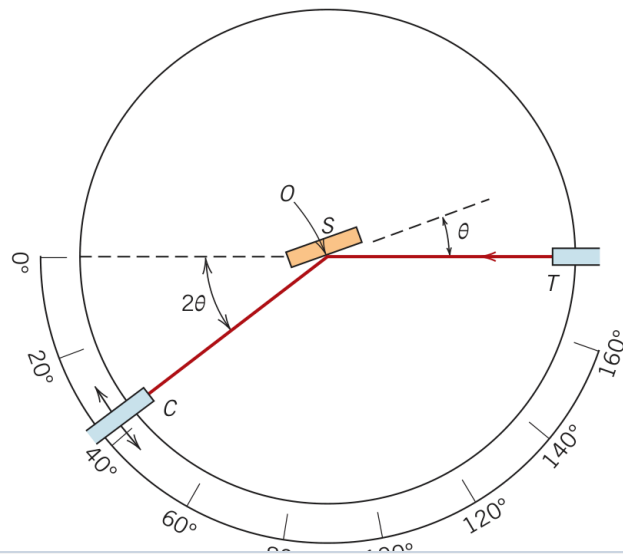


Figure 3.22: Operating geometry of an X-ray diffractometer showing source T , specimen S , and detector C rotating through angle 2θ .

In-Depth Visual Breakdown:

- **BCC Unit Cell (Figure 3.2):** Notice the hard-sphere contact. Atoms touch strictly along the cube **body diagonal** ($[111]$ direction). Therefore:

$$\text{Body Diagonal} = \sqrt{a^2 + a^2 + a^2} = a\sqrt{3} = 4R \implies a = \frac{4R}{\sqrt{3}}$$

- **X-Ray Diffractometer (Figure 3.22):** Monochromatic X-rays of known wavelength λ strike the powder specimen at angle θ . When the path difference between adjacent crystallographic planes equals an integer

number of wavelengths ($n\lambda$), constructive interference produces an intense diffraction peak recorded at detector angle 2θ .

4. Teacher Notes Cross-Reference & Concordia Exam Traps

- **Dr. Medraj Lectures 4–6 Emphasis:**
 - Step-by-step procedure for planes passing through the origin: **You must translate the origin** to an adjacent corner before identifying intercepts!
 - Close-packed stacking sequences: HCP is $ABABAB \dots$, while FCC is $ABCABCABC \dots$
 - Single crystals are **anisotropic** (properties like Young's modulus vary with crystallographic direction), whereas polycrystalline metals with random grain orientation are **isotropic** (quasi-isotropic) on a macro scale.
- **Concordia Exam Traps:**
 - **The θ vs. 2θ Trap in XRD:** Diffractometer outputs plot intensity against 2θ (the detector angle). You **must divide by 2** before plugging θ into Bragg's Law!
 - **Volume Unit Conversion:** Forgetting that $1 \text{ nm} = 10^{-7} \text{ cm} \implies 1 \text{ nm}^3 = 10^{-21} \text{ cm}^3$.

5. Master Exam Problem & Step-by-Step Solution Framework

Problem:

Iron has a BCC crystal structure with atomic radius $R = 0.1241 \text{ nm}$ and atomic weight $A = 55.85 \text{ g/mol}$. (a) Calculate its theoretical density. (b) For monochromatic X-radiation with $\lambda = 0.1542 \text{ nm}$, compute the diffraction angle 2θ for the first-order reflection ($n = 1$) from the (220) plane.

- **Step 1: Compute BCC Lattice Parameter a**

$$a = \frac{4R}{\sqrt{3}} = \frac{4(0.1241 \text{ nm})}{\sqrt{3}} = 0.2866 \text{ nm} = 2.866 \times 10^{-8} \text{ cm}$$

- **Step 2: Compute Unit Cell Volume V_c**

$$V_c = a^3 = (2.866 \times 10^{-8} \text{ cm})^3 = 2.354 \times 10^{-23} \text{ cm}^3$$

- **Step 3: Compute Theoretical Density ρ**

$$\rho = \frac{n \cdot A}{V_c \cdot N_A} = \frac{2 \times 55.85 \text{ g/mol}}{(2.354 \times 10^{-23} \text{ cm}^3)(6.022 \times 10^{23} \text{ mol}^{-1})} = 7.88 \text{ g/cm}^3$$

- **Step 4: Compute Interplanar Spacing d_{220}**

$$d_{220} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{0.2866 \text{ nm}}{\sqrt{2^2 + 2^2 + 0^2}} = \frac{0.2866}{\sqrt{8}} = 0.1013 \text{ nm}$$

- **Step 5: Apply Bragg's Law for $\sin \theta$ and 2θ**

$$\sin \theta = \frac{n\lambda}{2d_{220}} = \frac{1(0.1542 \text{ nm})}{2(0.1013 \text{ nm})} = 0.7611$$

$$\theta = \arcsin(0.7611) = 49.56^\circ \implies 2\theta = 99.12^\circ$$